

Supplementary Material for “Structural Entropy Dark Energy: a fixed-parameter, growth-gated holographic dark-energy model without a cosmological constant”

Stilian Pandev 

Independent researcher

E-mail: stilianpandev@gmail.com

Abstract. This supplement collects the supporting motivations and numerical probes for the volume-law horizon postulate, demoted from the main article to keep its main line focused: three further consistency motivations for the volume endpoint, the Majorana–SYK diagonalisation that fixes the horizon state, numerical probes of the state and counting halves of the postulate, the degrees-of-freedom accounting that locates the postulate as a counting (not thermalisation) claim — with the SYK and causal-set comparisons — and the summary of how far the postulate is derivable. None is a derivation; each is a consistency check. Cross-references of the form §N refer to sections of the main article; §S-N to sections of this supplement.

Contents

1	Supporting motivations and numerical probes for the volume-law postulate	1
1.1	Three further consistency motivations for the volume endpoint	1
1.2	The state is maximally entangled (Majorana-SYK diagonalisation)	2
1.3	The counting question — numerical probes (allowed, realisable, not selected)	2
2	The postulate is a dof-<i>counting</i> claim, not a thermalisation claim	3
3	Is the postulate derivable? Summary of the reduction (full treatment in the foundations companion paper)	4
	Code and data availability	5

1 Supporting motivations and numerical probes for the volume-law postulate

This supplement collects the material demoted from §8 of the main article to keep the main line focused: three further consistency *motivations* for the volume endpoint (§8.1) and the *numerical probes* of the state and counting halves of the postulate (§S2). None is a derivation; each is a consistency check that makes the postulate less *ad hoc* without selecting it. (The CHR near-critical layer, the cross-horizon black-hole reading, and the Verlinde dark-matter connection are developed in a separate foundations companion paper [1].)

1.1 Three further consistency motivations for the volume endpoint

(*supplementing the geometric ceiling, the CKN scale/form split, and the direct area-law disfavouring of §8.1*)

- **A driven steady state.** The horizon need not self-thermalise — its scrambling time ($\sim \ln S/H \approx 280/H$) far exceeds its age. Instead the structure-growth gate (the rise of f_{sat}) *drives* it off the ground state; with a scrambling-limited relaxation rate the volume-law state is *maintained* as a non-equilibrium steady state throughout structure formation (the long scrambling time becomes an asset). The volume law is then a fixed point of SEDE’s own dynamics, not an external input.
- *A roughening universality class* (**withdrawn**). This motivation modelled the horizon as a stochastic growing surface, $\partial h/\partial t = \nu \nabla^2 h + (\lambda_K/2)(\nabla h)^2 + \eta$, and argued the deformation was fixed by the roughening universality class. It is retracted: the count companion paper [2] shows the classical apparent horizon is constraint-slaved — $\theta_+ = 0$ is elliptic, with no growth equation at any order (its eq. 2.9) — and has no minimum-cut surface, so it does not roughen and there is no height field or roughening universality class. The ceiling $\Delta \leq 1$ and the value $\Delta = 1$ rest instead on the count paper’s leg-budget capacity theorem and on birth-selection (foundations §5), not on any surface-growth picture.
- An independent emergent-gravity proposal [3]. In Verlinde’s programme the de Sitter vacuum carries a volume-law contribution to its entanglement entropy, $S \propto V_H$, in addition to the horizon area law — the same super-area scaling SEDE adopts at $\Delta = 1$.

A second, independently-motivated route arriving at a volume-law entropy makes the postulate less *ad hoc*; it remains a motivation, not a proof, since Verlinde’s volume-law derivation is itself heuristic.

1.2 The state is maximally entangled (Majorana-SYK diagonalisation)

The *state* half of the postulate (§S2) — that the horizon dof are maximally entangled (thermal), not in a low-entanglement ground state — is settled and not peculiar to SEDE. Exact diagonalisation of the Majorana Sachdev–Ye–Kitaev model [4, 5], the canonical chaotic, black-hole-dual system saturating the Maldacena–Shenker–Stanford bound $\lambda_L \leq 2\pi T$ [6] (V60), confirms: the gap ratio reproduces Wigner–Dyson statistics with the correct N-mod-8 symmetry class [7–9], eigenstates saturate the Page value [10, 11], and the OTOC scrambles completely — while an integrable control does none of these. But this **does not fix** Δ : a black hole is a *maximal* scrambler yet area-law ($S = A/4$, reproduced by string microstate counts), and SYK, being all-to-all (geometry-free), cannot even pose the counting question.

(The SYK scrambling figure — the gap ratio climbing through the N-mod-8 RMT classes, the Page-value saturation of the eigenstate entanglement, and the complete OTOC scrambling of the $q=4$ model against the $q=2$ integrable control — is omitted from this version; the underlying diagnostics are the verification-suite check V60 and regenerate from the accompanying code.)

1.3 The counting question — numerical probes (allowed, realisable, not selected)

The *counting* half (§S2) — does the horizon dof number grow as area ($N \propto R^{d-2}$, $\Delta = 0$) or bulk volume ($N \propto R^{d-1}$, $\Delta = 1$)? — is where SEDE’s postulate lives. Three numerical probes (V62, V61, V63):

question	probe	result
Allowed? by the established entropy bounds	volume entropy vs the smooth-area bounds	$S \propto R^3$ overshoots the holographic/Bousso/Bekenstein bound $A/4$ by $\propto R$ ($\sim 10^{61}$ at the cosmic horizon); consistent only as a genuine $d_H = 3$ fractal horizon (true area $\propto R^3 \implies S = A_{\text{fractal}}/4$ <i>exactly</i> saturates the bound)
Selected? by quantum gravity	causal-set sprinkle, Minkowski and de Sitter	both scalings present; the canonical horizon object — causal <i>links</i> — is area (R^{d-2} , the Bekenstein–Hawking recovery), in flat space and, because dS is conformally flat, identically in de Sitter. Volume is not the default. (measured: bulk count $1.92 \approx 2 \implies \Delta = 1$; link count $0.98 \approx 1 \implies \Delta = 0$, in 2+1D)

question	probe	result
Realisable? in a holographic code	Ryu–Takayanagi min-cut, local vs nonlocal	local (lattice) connectivity $\implies$ area law (exponent 1.0); nonlocal (expander) connectivity $\implies$ volume law (exponent 1.9). $\Delta = 1$ is realisable; its required ingredient is <i>nonlocal</i> horizon connectivity (the all-to-all structure of SYK).

The pattern: volume-counting is allowed (not by evading the bounds but by the horizon being genuinely space-filling — the postulate itself), realisable (a nonlocal holographic code produces $S \propto R^{d-1}$ exactly), but not selected (the frameworks we can compute in default to area for the canonical horizon object, and a black hole is area-law). No consistency argument available to us *derives* the volume count; equally, none *excludes* it. The two deciders are stated in §S2: the dS static-patch Hilbert-space dimension (theoretical) and the deformation Δ (empirical, §5.6/§6).

2 The postulate is a dof-*counting* claim, not a thermalisation claim

The phrase “volume-law” conflates two claims of very different status. **(i) The state:** are the horizon degrees of freedom maximally entangled (thermal) or in a low-entanglement ground state? **(ii) The counting:** does the *number* of horizon dof grow as the area, $N \propto R^{d-2}$, or the spatial volume, $N \propto R^{d-1}$? Since $S \sim (\text{state factor}) \times N$, the deformation Δ — through $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$ — is fixed by the counting (area $\rightarrow \Delta = 0$, volume $\rightarrow \Delta = 1$ in $d=4$); the state only sets whether the prefactor is maximal. SEDE’s postulate is, precisely, the counting claim.

The state half is settled, and is not peculiar to SEDE. Maximal scrambling has a precise meaning in the Sachdev–Ye–Kitaev model [4, 5], the canonical chaotic, black-hole-dual system. But maximal scrambling **does not fix** Δ : a black hole is a *maximal* scrambler yet area-law ($S = A/4$, $\Delta = 0$, reproduced by string microstate counts). Scrambling is therefore shared with the $\Delta = 0$ black hole; and SYK, being all-to-all (geometry-free), cannot even pose the counting question. (An explicit Majorana-SYK diagonalisation confirming the state is maximally entangled — Wigner–Dyson statistics, Page-value saturation, complete OTOC scrambling — is given in §S1.2; it settles the *state*, not the counting.)

The counting half is where SEDE lives, and area is the default. In causal-set quantum gravity — the setting where the counting question is natural — both scalings appear, but the *canonical* horizon-entropy object, the causal-link count that recovers the Bekenstein–Hawking law (Sorkin and collaborators), is the area one (R^{d-2}), in flat space and identically in de Sitter. So even in the framework most hospitable to volume-counting, area-law is the default, and SEDE’s $\Delta = 1$ is the specific, non-default identification of horizon entropy with the bulk count. (The causal-set sprinkle measurements, the entropy-bound check, and a Ryu–Takayanagi holographic-code realisation — volume-law requires *nonlocal* horizon connectivity — are in §S1.3; their verdict is that the volume count is **allowed and realisable but not selected**: no consistency argument available to us derives it, and none excludes it.)

The sharpened postulate, properly located. The one open item is therefore *not* about thermalisation or chaos (settled, and shared with area-law black holes) but is purely a dof-counting statement:

the cosmic horizon’s entropy counts its bulk (volume-scaling, $N \propto R^3 \iff d_H = 3$) degrees of freedom rather than its boundary (area-scaling, links) ones — a count that exceeds the holographic area bound by R/ℓ_P (the seam of §8.3).

It reduces to two deciders. *Theoretically* it is precisely “*is the Hilbert-space dimension of the de Sitter static patch $e^{Area/4}$ or e^{Volume} ?*” — a frontier problem a complete dS holography would settle. *Empirically* it is the deformation Δ : $\Delta = 0$ (area) is already disfavoured by present data (§5.6), and DESI DR3 + Euclid will measure Δ to ~ 0.09 (§6). SEDE’s postulate is thus not an arbitrary assumption but a *consistent, realisable bet on a well-posed open question of quantum gravity*, with a decisive test imminent. Tellingly, the one horizon we can check today — the black hole — is area-law; on the evidence here this asymmetry is a genuine conceptual cost. The count companion paper [2] *offers* a resolution — a state-dependent count in which a black hole is area-law because *undriven* and only the structure-driven cosmic horizon activates the volume law — which, if its (conjectural) mechanism holds, would turn the cost into a predicted feature, with the black-hole side derivable and a sharp falsifier (an isolated volume-law black hole).

3 Is the postulate derivable? Summary of the reduction (full treatment in the foundations companion paper)

The volume-law postulate is not monolithic: it splits into a *state* (the horizon dof are maximally entangled/thermal), a *form* (the entropy scales as the volume), a *scale* (the magnitude is ρ_{crit} , not ρ_{Planck}), and a *count* (the number of dof grows as the bulk, $N \propto R^3$, not the boundary, $N \propto R^2$). The deformation Δ is fixed by the count alone. A systematic study of how derivable each piece is — five routes (de Sitter holography/SYK, entanglement first law in a thermal state, Verlinde emergent gravity, gravitational non-additivity, and a Bekenstein no-go), the reduction of the count to a *driven non-equilibrium steady state*, and three closure routes — is developed in the foundations companion paper [1]; we state the conclusions here, as they bear on how *ad hoc* the postulate is.

The *state* is first-principles (a maximal scrambler is volume-law-entangled; §S1.2, §S2); the *form* reduces to thermalisation (volume-law entanglement is the generic behaviour of a thermal reference state for regions larger than the thermal length); the *scale* is CKN (§8.2). What remains irreducible is purely the count, and a Bekenstein no-go makes precise *why*: maximum-entropy/energy bounds give the *area* law (§8.3), so the volume count cannot be obtained from any energy argument — it is a genuine state/counting statement. The count is, in turn, downstream of *connectivity* (locally-connected horizon dof give area-law entropy, nonlocal give volume-law) and connectivity of *interaction range*: gravity is strongly long-range ($1/r$ with $\alpha = 1 \leq d$), hence non-additive — the volume class is the *natural* one for a self-gravitating horizon, with area-law arising only when the holographic bound intervenes for an *isolated, equilibrium* horizon (a black hole). The cosmic horizon differs in being continuously *driven* by structure formation, which converts the black-hole/cosmic asymmetry (§7) into a discriminator (equilibrium $\rightarrow$ area, driven $\rightarrow$ volume) and reduces the residue to a single, falsifiable **driven-NESS statement**: a strongly-long-range horizon, driven by structure formation, settles in the volume-law steady state rather than relaxing to the area-law

equilibrium. The foundations companion paper supplies the two mechanisms this requires — *selection at the horizon’s birth* (the N-dependent cooperative coupling $J/J_c = 2\pi Gm^2N$ carries the horizon, grown from Planckian size, through the ordering bifurcation, with the Barrow tilt selecting the volume branch and long-range hysteresis locking it; no barrier is ever crossed) and the structure *drive*, whose envelope is the gate f_{sat} , which *populates* the selected volume capacity rather than selecting the branch — so the static *counting* postulate is replaced by the weaker, more physical statement of a birth-selected branch plus derived occupancy kinetics.

Closure — deriving even that bistable free energy from first principles — would reduce the dark sector to Ω_m plus established physics. It takes three precise forms: a de Sitter-holography computation of the bulk count (open, and provably not shortcut-able through energy bounds); exhibiting the volume branch as a real saddle/state (the bistable landscape follows from gravitational cooperativity given that branch); or *measuring* Δ . The last is decisive and already in hand: DESI DR3 + Euclid pin Δ to ~ 0.09 (§6), so $\Delta = 1$ establishes the volume count as fact and $\Delta = 0$ refutes it. The postulate is thus a single, sharply-posed, falsifiable statement with an imminent test — not an unconstrained assumption.

Code and data availability

All code and analysis pipelines are publicly available at <https://github.com/spsingularity/sede-cosmology>; the numerical diagnostics above are regenerated by the accompanying code. A tagged release is archived at Zenodo, DOI [10.5281/zenodo.21050314](https://doi.org/10.5281/zenodo.21050314).

References

- [1] S. Pandev, “Is the volume-law horizon a postulate or a consequence? Birth-selection and a driven non-equilibrium reformulation of Structural Entropy Dark Energy.” 2026.
- [2] S. Pandev, “The horizon count from a leg-budgeted long-range network.” 2026.
- [3] E.P. Verlinde, *Emergent Gravity and the Dark Universe*, *SciPost Phys.* **2** (2017) 016 [[1611.02269](#)].
- [4] S. Sachdev and J. Ye, *Gapless spin fluid ground state in a random, quantum Heisenberg magnet*, *Phys. Rev. Lett.* **70** (1993) 3339 [[cond-mat/9212030](#)].
- [5] A. Kitaev, “A simple model of quantum holography.” Talks at KITP, Entanglement in Strongly-Correlated Quantum Matter program, Univ. of California, Santa Barbara, 2015.
- [6] J. Maldacena, S.H. Shenker and D. Stanford, *A bound on chaos*, *JHEP* **08** (2016) 106 [[1503.01409](#)].
- [7] A.M. García-García and J.J.M. Verbaarschot, *Spectral and thermodynamic properties of the Sachdev-Ye-Kitaev model*, *Phys. Rev. D* **94** (2016) 126010 [[1610.03816](#)].
- [8] J.S. Cotler, G. Gur-Ari, M. Hanada, J. Polchinski, P. Saad, S.H. Shenker et al., *Black Holes and Random Matrices*, *JHEP* **05** (2017) 118 [[1611.04650](#)].
- [9] Y.Y. Atas, E. Bogomolny, O. Giraud and G. Roux, *Distribution of the Ratio of Consecutive Level Spacings in Random Matrix Ensembles*, *Phys. Rev. Lett.* **110** (2013) 084101 [[1212.5611](#)].
- [10] D.N. Page, *Average entropy of a subsystem*, *Phys. Rev. Lett.* **71** (1993) 1291 [[gr-qc/9305007](#)].
- [11] L. Vidmar and M. Rigol, *Entanglement Entropy of Eigenstates of Quantum Chaotic Hamiltonians*, *Phys. Rev. Lett.* **119** (2017) 220603 [[1708.08453](#)].